

Pranay Kashyap

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EDUCATION

- **New York University**
• *Master of Science in Applied Data Science*
- **National Institute of Technology**
• *Bachelor of Technology in Information Technology*

Brooklyn, NY
Sep. 2024 – May. 2026
Srinagar, India
Aug. 2018 – Jul. 2022

SKILLS

- **Languages:** Python, Java, JavaScript, Kotlin, C/C++, SQL
- **ML/AI:** PyTorch, TensorFlow, Keras, Gen AI, Video AI, Autoregressive Modeling, Representation Learning, Computer Vision, OpenCV, ONNX, NLP, LLMs
- **Tools & Platforms:** FastAPI, Flask, Docker, Triton Server, WebSockets, CI/CD, AWS, FFmpeg, Android SDK/NDK, Jetpack, Firebase, Gradle, Git, CUDA, Anaconda, Linux

EXPERIENCE

- **Kaliber AI** San Mateo, CA
Machine Learning Engineer Jun 2026 - Present
 - Developing a multimodal predictive world model combining frozen V-JEPA ViT-L representations with 3D state history to learn short- and long-horizon dynamics, forecasting future 3D position and velocity states from 32-frame visual context.
 - Trained an autoregressive trajectory predictor with closed-loop rollout and backpropagation through time, recursively forecasting future 3D states from predicted V-JEPA latents to mitigate long-horizon error accumulation and distribution shift.
 - Architected and trained from scratch a convolutional latent-to-video decoder, progressively upsampling autoregressive V-JEPA latents to generate 60 future frames with disentangled background/object RGB and alpha rendering, coupled with physics-aware motion and temporal-consistency objectives.
 - Developed a framework to interpret and steer LLM conversational trajectories, probing layer-wise residual-stream activations to decode current and future discourse states and trajectory dynamics, quantify attractor basins and representation drift, and learn activation-level controllers that steer models toward target conversational states while preserving behavioral traits.
- **Kaliber AI** San Mateo, CA
Machine Learning Intern May 2025 - Dec 2025
 - Developed a real-time medical voice assistant with a modular FastAPI backend, WebSocket streaming, speaker-verified Voice Activity Detection, and optimized Whisper-large-v3, enabling partial-text transcription and achieving sub-2s latency.
 - Engineered interrupt-driven TTS and dynamic buffering to reduce perceived latency by 35% and improve conversational flow.
 - Containerized and deployed full AI stack on AWS EC2 (g5.xlarge) and Triton Inference Server, leveraging GPU acceleration, TensorRT-optimized models, and Dockerized microservices for reproducible, zero-downtime production deployments.
- **Samsung Research** Bengaluru, KA
Senior Software Engineer Jul 2022 - Aug 2024
 - Led the implementation of a DNN model for Generic Event Boundary Detection (GEBD), now a core component of the Video Moments solution in the Enhance-X app. This app, available on the Galaxy Store, processed data on-device in real-time, helping users efficiently identify and manage key moments in their videos.
 - Boosted accuracy in video event captioning by 20% by modifying CLIP to incorporate optical flow and pixel movement, capturing temporal dynamics for precise boundary-aware captions.
 - Developed a zero-cost approach for GEBD using the Segment Anything Model (SAM), achieving state-of-the-art results, and publishing the work at ICASSP'24, this method repurposed SAM for class-agnostic video boundary detection.
 - Optimized GEBD model through ONNX-based quantization, cutting size by 71.4% and accelerating edge deployment with sustained KPI performance and reduced computational overhead.
 - Enhanced backend performance by integrating scalable APIs using FFmpeg for real-time video frame processing.
- **Samsung Research** Bengaluru, KA
Software Engineer Intern Jan 2022 - Jul 2022
 - Streamlined GEBD by retraining BSN and BMN models, cutting computational complexity by 80%.
 - Developed a self-similarity-based 3D filter pruning method to compress CNNs without post-training finetuning, achieving 12% faster inference and 10% model size reduction on video models, enabling deployment on edge devices.
 - Applied 3D filter pruning to X3D in the GEBD pipeline, cutting size and speeding inference, with near-zero accuracy drop.

RESEARCH PAPERS & PATENTS

- * SAMnet: Semantic Aware Multimodal Network for Emoji Drawing Classification — CVIP'22
- * Repetition Counting from Compressed Videos Using Sparse Residual Similarity — ICASSP'23
- * Self-Similarity is All You Need for Fast and Lightweight GEBD — ICASSP'23
- * Whats in the Flow Exploiting Temporal Motion Cues for Unsupervised GEBD — WACV'24
- * SAM-GEBD: Zero-Cost Approach for Generic Event Boundary Detection — ICASSP'24
- * Method and electronic device for generating a segment of a video (2023)
- * Generating contextual transition effects in videos (2024)
- * Electronic device and method for adaptive video display management based on region localized regeneration of frames (2024)

AWARDS

- * **LM-Studio On-Device AI Builders Hackathon'24** : Built an on-device app using Donut and LLaMA models.
- * **Samsung Best Paper Award 2024 - Bronze** : Awarded for paper on physics-based interactive dynamics.
- * **Samsung IFest'23 Ideation Contest** : For creating breakthrough idea and showcasing PoC on VideoAI.